

Modelling Traffic Policy with Drivers and Government Behavior using Evolutionary Game Theory

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Abstract. The article proposes a model that describes the dynamics of offenses in terms of non-compliance with safety regulations and speeding, taking into account established rules and national specifics. The method discussed in the article is a combination of game theory methods and evolutionary algorithms, where "soft" and "strict" government policies compete with each other in a population of states, and law-abiding and law-breaking drivers compete in a population of authors. The study examines the sensitivity of drivers to sudden changes in government policies and the impact of penalties on driver behavior within the model.

Keywords: Evolutionary game theory, government regulation, traffic policy, speeding rate, government spendings, seatbelt wearing rate

1 Introduction

Road safety has become one of the main state tasks over the past hundred years. The key factors in increasing the importance of the issue of traffic control were, first of all, the increase in the number of drivers, advances and limitations in technologies, as well as the degree of interest of road policy developers in increasing the mobility of the population.

First of all, the states are interested in minimizing road accidents, largely due to the fact that the latter cause significant harm to GDP. For example, at the time of 2021, the International Transport Forum report on Germany claimed that road accidents cost the state treasury about 0.9% GDP [1].

Using Europe as an example we can observe a positive trend in road safety, which expressed a significant reduction in the number of fatal accidents [2], however the issue of implementing policies that have reduced the frequency of accidents is still acute for all states, as they cause cumulative economic damage [3] summarizing the cost of repairing damaged vehicles, losses caused by transport downtime, the cost of reconstructing and repairing damaged roads and road structures, the cost of investigating a traffic accident by police officers, etc.

Due to these and other reasons, the issue of modeling road safety as a system that takes into account various forms of regulation and the social context is still relevant.

Within the framework of the study under consideration, we set ourselves the task of creating an environment model which made it possible to describe the dynamics of

offenses in terms of non-compliance with the speed limit and wearing a seat belt, taking into account the socio-psychological specifics of a certain national environment (for example, a community or a state), including regulatory specifics.

Conducting numerical modeling using the example of Germany, we will consider the general dynamics of law enforcement in the context of two different government policies for regulating the road sector, expressed at the initial stage in the state of the road infrastructure, fixed fines for violations of the speed limit and an unbuckled seat belt, with fixed parameters of national specifics and driver behavior patterns.

The study will also address the issue of drivers' resilience to a sharp tightening or easing of state regulation of road transport, consider the impact of certain system parameters on the frequency of offenses and overall government spending, and explore the importance of penalty policies on changing the dynamics of offenses among drivers.

2 Related work

At the moment, the topic of modeling legal regulation with an account of socio-psychological context is poorly searched. Most of the researches focuses either on strategies for organizing police to minimize accidents [4], or on modeling the driver's decision-making function [5], or specific traffic situations are studied using the example of games [6], however, very few studies link the forms of regulation and their impact on the dynamics of offenses.

We were also interested in an article that discussed agent-based transportation modeling using government control standards [7]. At the same time, research is mainly aimed at studying the issue of regulation of commercial transportation, but not private transport.

The work was also based on a review study [6, 8] which analyzed the usage of game theory elements to methods various traffic situations, for example, the confrontation between drivers exceeding the speed limit and the police.

In addition, the work is based on studies of the driver's satisfaction with the trip [9], which revealed that one of the main factors is the duration of the latter.

The reports of the European commissions and public organizations [10, 11, 14, 15] and the report submitted within the framework of the international transport forum [1], which allowed to initialize the model parameters, also played a key role in the study.

3 Traffic policy modeling with evolutionary game theory

The basis for our Agent-based model is a combination of two theories: the theory of coevolutionary algorithms and the theory of evolutionary games. In our model, within the framework of the general environment, there will be many agents of the "state" class who implement a certain traffic regulation policy, and many agents of the "driver" class who can violate or observe the speed limit, as well as drive with or without a seat belt.

A model of the dynamics of law enforcement in the road sector (see Fig. 1) was based on the assumption that all drivers in the system drive identical cars, the police are

always called to the scene in case of an accident, and drivers wait for their arrival in case of the last one.

The model does not take into account the progressive penalty rate. Also, to simplify the construction of the model, we will consider that license freezing is not to be used by the state as a tool to punish the driver for exceeding the speed limit. In addition, we will assume that in order to impose a fine, the police must stop the driver.

Based on the results of our research in the literature, we decided to build a model based on the assumption that the key factor in driver's satisfaction with the road is the subjective time spent on the trip [9]. We also believe that the condition of the road affects the frequency of accidents [13].

For states, the key indicator of the success of the implemented policy will be the average cost of the implemented policy per driver including fees and crashes economic losses.

Within the framework of the model, we assume that the country we are considering has a federal structure, which makes it possible to implement different regulatory measures in different regions. This assumption will allow tracking the overall dynamics of offenses, as well as budget expenditures.

Since the chance of a fatal accident is lower than the chance of a non-fatal accident, the cost of an accident will be calculated without taking into account fatal outcomes.

Finally, we assume that the time of arrival of the police at the scene of the accident is a random value, but the duration of the inspection is a constant value, regardless of whether it is carried out after.

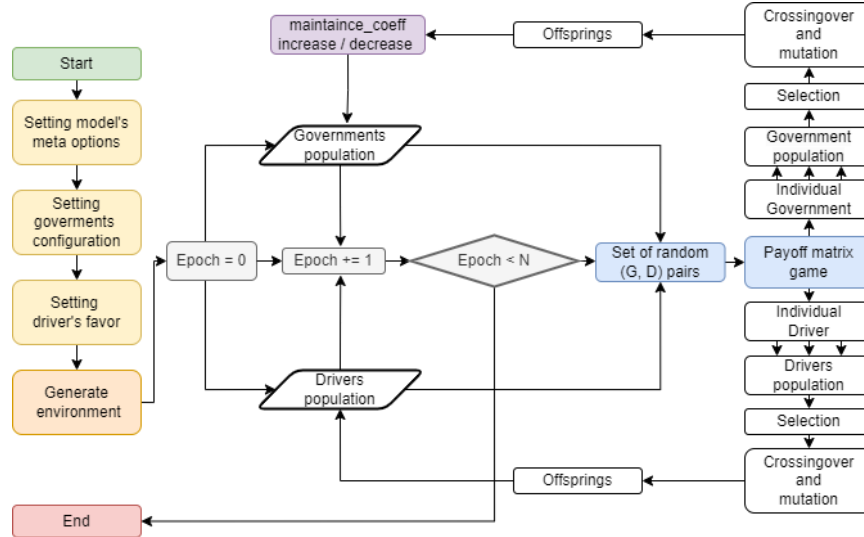


Fig. 1. The scheme describes the algorithm of the system for conducting a single experiment.

In the following subsections, let's take a closer look at the most important stages of the experiment model.

3.1 Initialization

At first step the parameters defining the environment are initialized. These include the parameters of national specifics (see Table 1), the meta-parameters of the system (see Table 2), the parameters of driver behavior (see Table 3) and the parameters of strict and lite government policies implemented within a single environment (see Table 4). Since the set of parameters for the two policies is identical, in Table 4 the latter are presented in a general form without reference to strict or mild form of regulation.

Table 1. Parameters of regulatory policies in the state.

Name	Unit	Norway	Germany	Meaning
crash_chance	prob	0.0203	0.051	The chance of an accident in the state
fatal_chance	prob	0.00002	0.00005	The chance of a fatal accident in the state
mu_wait (mu)	hours	0.27	0.27	The average waiting time for police to arrive at the scene of an incident in the state
sigma_wait	hours	0.05	0.05	The average square deviation of the police arrival time at the scene in the State
check_duration	hours	0.25	0.25	The time of checking the car in the state
check_chance	Prov	0.6	0.4	The parameter determines the chance of checking the driver in the state
traffic_spendings	hours	428	373.8	The average amount of expenses per driver for the implementation of state policy
fee_belt_min	Euro	132	10	The left boundary of the uniform distribution of fines for unbuckled drivers in the state
fee_belt_max	Euro	264	20	The right boundary of the uniform distribution of fines for unbuckled drivers in the state
fee_s_min	Euro	100	60	The left boundary of the uniform distribution of fines for drivers exceeding the speed limit in the state
fee_s_max	Euro	1410	400	The right boundary of the uniform distribution of fines for drivers exceeding the speed limit in the state
maintance_coeff (maint)	-	0.8	0.6	The initial value of the coefficient of the infrastructure of the state (takes values from 0 to 1)
lite_g_chance	prob	0.05	0.05	The chance of a soft-regulated state at the time of mutation
strict_g_chance	prob	0.05	0.05	The chance of a tightly regulated state at the time of mutation

In the future, to detail the parameters of regulation between strict and lite policies, "_s" and "_l" will be added to the names of characteristics, respectively. For example, `mu_wait_l` is a parameter applicable to a state with a lite regulatory policy.

Table 2. Meta parameters

Name	Unit	Value	Meaning
N_runs	uint	10	Number of experiments
N_epochs	uint	300	Number of epochs in one experiment
N_drivers	uint	1000	Number of drivers in the experiment
crash_cost	euro	137.008	Average cost of an accident
fatal_belt_min (fatal_min)	-	0.6	The coefficient of reduction of fatal outcome when driving with a belt
delta_maintenance	-	0.05	A parameter characterizing the rate of degradation of an infrastructure
chance_of_mutation	-	0.05	The probability of driver/state agent mutation

After setting the parameters, the system proceeds to generate an environment in which many agents of the «driver» type and many agents of the «state» type (each of the N_drivers dimension) are created, after which the system conducts a series of games.

Table 3. Parameters of national specifics

Name	Unit	Value	Meaning
decrease_coeff	-	0.8	Reducing the coefficient of subjective perception of time spent on the road when speeding
fatal_coeff	-	100	An increasing coefficient that increases the chance of a fatal outcome in case of violation of the speed limit
crash_coeff	-	1.5	An increasing coefficient that increases the chance of an accident in violation of the speed limit
spendings_coeff	-	0.8	The coefficient of subjective perception of expenses incurred by the state on average
fee_tolerance_coeff	-	0.8	A reducing coefficient that characterizes the degree to which drivers pay their fines.
check_coef	-	60	The coefficient of subjective perception of the check time by the driver

Each agent of the driver type is generated independently of each other. The driver is equally likely to be assigned one of two driving strategies (observing or violating the speed limit) and, regardless of this, the belt status (fastened or not). The driver also generates the value of the expected travel time, as well as a random speed from the range from *speed_n_min* to *speed_n_max*, if the driver adheres to the speed limit, and from the range from *speed_s_min* to *speed_s_max*, if not.

Each agent of the "state" type is generated independently of each other. With equal probability, the state is assigned one of two regulatory strategies («lite» or «strict»), after which the latter is assigned a random belt penalty from a uniform distribution with boundaries between *fee_belt_min* and *fee_belt_max* and for speed, and a random fine from a uniform distribution with boundaries between *fee_s_min* and *fee_s_max*. The coefficients are selected depending on the previously selected strategy. Also, taking into account the control strategy, the constant policy characteristics are assigned to the agent of the "State" type: *crash_chance*, *mu_wait*, *sigma_wait*, *check_duration*, *check_chance*, *traffic_spendings*, *maintance_coeff*, *fatal_chance*.

Table 4. General parameters of driver behavior.

Name	Unit	Value	Meaning
speed_n_min	km/h	70	The left boundary of the uniform speed distribution for drivers who follow the speed limit
speed_n_max	km/h	100	The right boundary of the uniform speed distribution for drivers who follow the speed limit
speed_s_min	km/h	110	The left boundary of the uniform speed distribution for drivers exceeding the speed limit
speed_s_max	km/h	150	The right boundary of the uniform speed distribution for drivers exceeding the speed limit
exp_road_duration (exp)	hours	1	The parameter determines the subjective average expected travel time for the driver.
sigma_road_duration (sigma)	hours	0.1	The parameter determines the average square deviation of subjective travel time from the driver's perspective.
length	km	100	The actual length of the path

3.2 Series of games

To conduct a series of games randomly, each element from the set of agents of the "Driver" type is assigned an element from the set of agents of the "State" type. Each pair of «Driver»-«State» agents participates in the game described by the payoff matrix (see Table 5). Note that all random components drop out once per pair

Table 5. Payoff matrix for games

Government's Regulation strategy	Driver's speed strategy	
	Obeying speed limit	Speeding
Strict regulation	GSO, DSO	GSS, DSS
Lite regulation	GLO, DLO	GLS, DLS

Let's write out the formulas for calculating score for the state and drivers for each strategy case. Let's start with the driver, who obey the speed limit in strict government:

$$DSO = N(\exp, \sigma^2) - SOSpeed - SOFatal - SOCrash - SOCheck$$

$$SOSpeed = \frac{length}{speed \cdot maint}$$

$$SOFatal = Bern(fatal_chance_s \cdot (1 - belted \cdot (1 - fatal_min))) \cdot 10^6$$

$$SOCrash = Bern(crash_chance_s) \cdot SODur$$

$$SODur = N(mu_s, sigma_wait_s^2) + check_duration_s \cdot check_coef$$

$$SOCheck = Bern(check_chance_s) \cdot check_duration_s \cdot check_coef$$

Now let's set the score for the driver, who obey the speed limit in the lite government. The only difference from the previous case is the softer policy coefficients:

$$DLO = N(exp, sigma^2) - LOSpeed - LOFatal - LOCrash - LOCheck$$

$$LOSpeed = \frac{length}{speed \cdot maint}$$

$$LOFatal = Bern(fatal_chance_l \cdot (1 - belted \cdot (1 - fatal_min))) \cdot 10^6$$

$$LOCrash = Bern(crash_chance_l) \cdot LODur$$

$$LODur = N(mu_l, sigma_wait_l^2) + check_duration_l \cdot check_coef$$

$$LOCheck = Bern(check_chance_l) \cdot check_duration_l \cdot check_coef$$

Now let's set a score for a driver who exceeds the speed limit in a strict state. This driver will be affected by the road conditions, but a subjective coefficient will be applied based on the increase in speed:

$$DSS = N(exp, sigma^2) - SSSpeed - SSFatal - SSCrash - SSCheck$$

$$SSSpeed = \frac{length}{speed \cdot maint} \cdot decrease_coeff$$

$$SSFatal = Bern(fatal_chance_s \cdot fatal_coeff \cdot (1 - belted \cdot (1 - fatal_min))) \cdot 10^6$$

$$SSCrash = Bern(crash_chance_s \cdot crash_coeff) \cdot SSDur$$

$$SSDur = N(mu_s, sigma_wait_s^2) + check_duration_s \cdot check_coef$$

$$SSCheck = Bern(check_chance_s) \cdot check_duration_s \cdot check_coef$$

Finally, let's set score for the driver, who is speeding in government with lite regulation. Again, the only difference from the previous case is the softer policy coefficients:

$$DLS = N(exp, sigma^2) - LSSpeed - LSFatal - LSCrash - LSCheck$$

$$LSSpeed = \frac{length}{speed \cdot maint} \cdot decrease_coeff$$

$$LSFatal = Bern(fatal_chance_l \cdot fatal_coeff \cdot (1 - belted \cdot (1 - fatal_min))) \cdot 10^6$$

$$LSCrash = Bern(crash_chance_l \cdot crash_coeff) \cdot LSCheckDur$$

$$LSCheckDur = N(\mu_l, \sigma_l^2) + check_duration_l \cdot check_coef$$

$$LSCheck = Bern(check_chance_l) \cdot check_duration_l \cdot check_coef$$

Note that the *beltd* is a parameter which identifies is seat belt fastened or not for a specific agent of the "Driver" type (1 – the driver is fastened, 0 – the driver is not fastened). Under $N(\mu, \sigma^2)$, we consider a random variable distributed according to the normal distribution law with parameters μ and σ^2 . By $Bern(p)$, we consider a random variable distributed according to the Bernoulli distribution law with the parameter p . *speed* is the value of the speed parameter of the specific driver.

Next, we will take similar steps for the states. First, we will set the scores for governments with strict policy in scenario when such a state encounters a law-obeying driver:

$$GSO = SOSpend + beltd \cdot fee_belt$$

$$SOSpend = (-SOCost - traffic_spendings_s) \cdot spendings_coeff$$

$$SOCost = crash_cost \cdot Bern(crash_chance_s)$$

Now, let's consider scenario when the state encounters a speeding driver:

$$GSS = SSSpend + beltd \cdot fee_belt + fee_speeding$$

$$SSSpend = (-SSCost - traffic_spendings_s) \cdot spendings_coeff$$

$$SSCost = crash_cost \cdot Bern(crash_chance_s \cdot crash_coeff)$$

Next let's consider that governments with lite policy faces with a law-abiding driver:

$$GLO = LOSpend + beltd \cdot fee_belt$$

$$LOSpent = (-LOCost - traffic_spendings_l) \cdot spendings_coeff$$

$$LOCost = crash_cost \cdot Bern(crash_chance_l)$$

Finally, let's set the score function for the governments with lite policy which faces with speeding driver:

$$GLS = LSSpend + beltd \cdot fee_belt + fee_speeding$$

$$LSSpend = (-SSCost - traffic_spendings_l) \cdot spendings_coeff$$

$$LSCost = crash_cost \cdot Bern(crash_chance_l \cdot crash_coeff)$$

3.3 Selection

In a series of $N_drivers$ games, each agent of the sets of "States" and "Drivers" receives a reward. The system then generates subsets for each of the agent types, consisting of half of the initial elements with the highest reward rates, after which it transmits the received arrays to the crossing over.

3.4 Crossing over

As in the previous paragraph, the crossing is carried out independently for many drivers and policies separately. Due to the differences in the algorithms, we will consider each of them separately.

The resulting set of drivers is duplicated and «Driver»-«Driver» pairs are randomly formed, in each of which there is an equal probability of either an exchange of the *belted* feature, or a proportional exchange of speeds, or both an exchange of the *belted* feature and a proportional exchange of speeds, after which a new unified population is formed from the two subsets, which is transmitted by mutation

The resulting set of policies is duplicated and «State»-«State» pairs are randomly formed, each of which is equally likely to have either an exchange of the *maintance_coeff* feature, or a proportional exchange of penalty policies for belts, or a proportional exchange of penalty policies for exceeding the speed limit, after which a new single population is formed from the two subsets, which is transmitted by mutation.

3.5 Mutation

As in the previous paragraph, mutation is performed independently for multiple drivers and policies separately. Due to the differences in algorithms, we will consider each of them separately.

For each element of the driver set, a new speed for the current behavior strategy can be generated with a chance equal to *chance_of_mutation*, the "belted" feature is replaced with the opposite, and a new agent is created to replace the previous one from the opposite strategy class, but with the same "belted" feature. This results in a new population sample and the end of the driver era.

For each element of the policy set for the current implemented policy, a new penalty for violating safety rules for the current behavior strategy can be generated with a chance equal to *chance_of_mutation*, a new penalty for speeding for the current behavior strategy can be generated, the *maintance_coeff* attribute can be replaced with a random value from 0.4 to 1, and a new agent can be created to replace the previous strategy from the opposite class with a random set of characteristics.

After mutation, for all agents with a «Strict» management policy, the *maintance_coeff* indicator increases by *delta_maintance*, if it does not exceed the threshold of 1, for agents with a «Lite» regulation policy, the *maintance_coeff* indicator drops by *delta_maintance* to at least 0.4. After that, a new population sample is formed and the epoch for repeated states ends.

Upon completion of mutations, the epoch counter increases by 1.

3.6 Exit condition

The experiment ends when the epoch variable reaches the parameter value *N_epochs*. After that, the system completes the experiment.

4 Case study: modeling of traffic policy environment in Germany

In this section, we will conduct a series of experiments aimed at studying the general dynamics of incidents and the influence of factors on key indicators.

First, let's look at some examples. As part of the study, we will examine the dynamics of incidents using the example of two European countries, since countries in it had a common historical and social development background, but, on the other hand, all of them still have differences in national specifics of regulation. Meanwhile, over the past few years, Norway has been leading among all European countries in road safety, imposing the highest fines for non-compliance with road rules in the entire European Union. For this reason, we will consider the Norwegian security control strategy to be "strict". To some extent, we can consider the Norwegian strategy to be ideal, compared to which all other policies can be considered lite. Due to the fact that Germany has a lower road safety rating [1, 2], and its penalty policy is more lenient than the norm, we will consider its control strategy lite.

Let's define the initialization parameters of government strategies (see Table 6). The chances of accidents and fatalities were taken from the reports on Germany [14]. For Norway, the values were calculated proportionally. For the sake of simplicity, we assumed that the μ and σ_{wait} parameters of the two states were the same, since we did not find any feedback. For the same reason, the values of the parameters were taken from American studies [12]. The duration of the check also matches. Also, for simplicity, we assigned the parameters $maint_s$, $maint_l$, $check_chance_s$, $check_chance_l$ the values 0.8, 0.6, 0.6, 0.4 based on the ranking of the German and Norwegian models by safety [2]. The boundaries of penalty policies for exceeding the speed limit were defined as a minimum fine for exceeding the limit to 5 km/h and a maximum fine for exceeding the limit to 50 km/h. The left boundary of fines for not wearing a belt was defined as the current state fine, and the right boundary was the current fine, doubled. Note that for Norway fines were converted from NOK to euros at the exchange rate current at the time of February 2026.

Since we could not find information in open sources on how much the government of Germany and Norway spends on road safety, the average spending on policy implementation was calculated as the sum of government expenditures on Police services and R&D Public order and safety in euros taken from the EU report for 2023 [15] divided by the total number of the population of the country at the time of 2023.

As part of the experiment, we simulate the case when drivers drive in rural areas where speed limits in Germany and Norway allow acceleration to 100 km/h. In experiment we set speed initialization limits between 70 km/h and 100 km/h for drivers who follow the speed limit, and between 100 km/h and 150 km/h for those who do not. We do not consider higher speeds, because for exceeding the speed limit above 50 km/h, the driver is deprived of his license. The trip length is set at 100 km, the average expected travel time is 1 hour with an average deviation of 0.1 h.

Finally, assuming that the average cost of an accident is the same for all countries and the latter includes only cases with severe and mild injuries (without fatal outcomes,

since they are quite rare), the cost of an accident was taken from reports for Norway and amounted to 1.539.328 NOK or 137.008 euro [11].

Since the model has a pronounced stochastic nature, we set 10 launches with 300 epochs per launch in one experiment. *Delta_maintenance* was determined to be 0.05. The *fatal_min* indicator will be 0.6. The parameters of Germany's national specifics (see Table 2) were selected through a series of launches with an attempt to approximate the values of indicators expressed in the percentage of drivers who do not use a seat belt, which is 98%, and in the percentage of drivers who exceed the speed limit, which is 74% [10]. With the current parameters for an average of 300 epochs, taking into account the averaging of trajectories over 10 launches, 90.4% of drivers wear seat belts (with an average deviation of 4.9%), and 74.6% of drivers exceed the speed limit (with an average deviation of 4.9%).

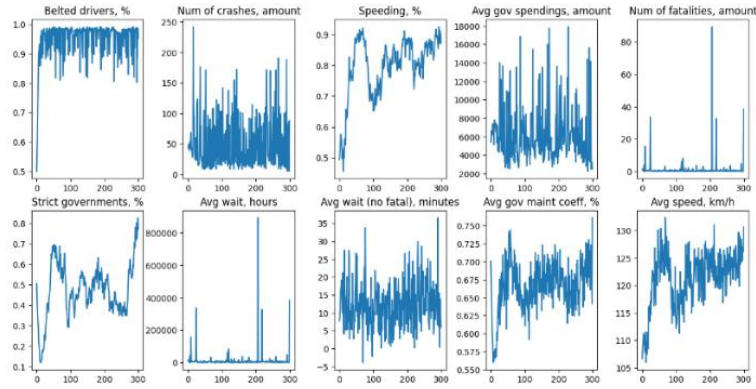


Fig. 2. Dynamic graphs when running on fixed parameters without modifications

4.1 Two-way ANOVA for belt and speeding fees

Let's look at the impact of the right-hand boundaries of penalty policies on the dynamics of violations of laws regarding the speed limit and the presence of a seat belt by conducting a series of two two-way ANOVA tests. For one case, we fix all the parameters except for pairs of values *fee_belt_max_s* - *fee_belt_max_l* (set between 600 and 2200 with step 400 and between 200 and 600 with step 100, respectively) and *fee_s_max_s* - *fee_s_max_l* in the other (set between 200 and 400 with step 50 and between 50 and 200 with step 50, respectively). All other parameters remain fixed. After that 2 series of experiments are launched, each with 25 independent experiments on all a grid of 5 for each of the modifiable parameters, fixing the average values of the percent of buckled and speeding drivers in each of the 10 iterations of the experiment. Based on the collected data, a Two-way ANOVA was performed (see fig. 3 and fig. 4), which showed that the factors *fee_belt_max_s*, *fee_belt_max_l* and their combination do not affect the frequency of violations. At the same time factor *fee_s_max_l* effects on speeding with p-value of 5%, but there is no influence of the *fee_s_max_s* factor and a combination of factors.

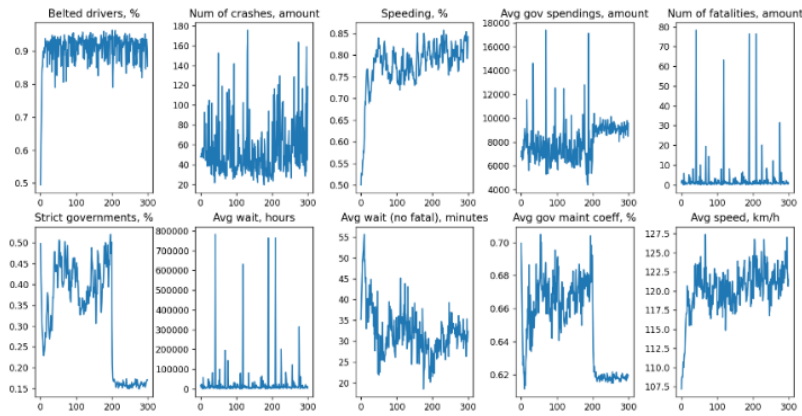
Table 4. two-way ANOVA tests results.

Factor name	P(>F)	
	Influence on ratio of belt usage	Influence on ratio of speeding
$C(\text{fee_belt_max_l})$	0.467	0.8985
$C(\text{fee_belt_max_s})$	0.84043	0.6211
$C(\text{fee_belt_max_l}):C(\text{fee_belt_max_s})$	0.6738	0.1159
$C(\text{fee_s_max_l})$	0.1917	0.011*
$C(\text{fee_s_max_s})$	0.7595	0.521
$C(\text{fee_s_max_l}):C(\text{fee_s_max_s})$	0.1093	0.674

* $p < 0.05$

4.2 Sensitivity test

In this section, we will conduct a visual study of drivers' resilience to drastic changes in government policies. To begin with, let's run a regular experiment (see Figure 2). Based on the latter, we will construct averaged plots of key characteristics, which include: the ratio of speeding drivers, the average number of accidents per era, the ratio of speeding to the total number of drivers, the average government expenditure per era, the number of fatalities, the relative number of strict government policies to the total number of policies, average time spent traveling above the scheduled time for driver, average time spent traveling above the scheduled time for driver (excluding fatal accidents), the average value of the infrastructure indicator and the average speed indicator. Next, we will conduct two more runs. This time, after the two-hundredth epoch, in one case, $\text{lite_g_chance} = 0.75$ (see Figure 5), and in the other case, $\text{strict_g_chance} = 0.75$ (see Figure 6).

**Fig. 5.** Dynamics graphs running with $\text{lite_g_chance} = 0.75$ after the 200th epoch

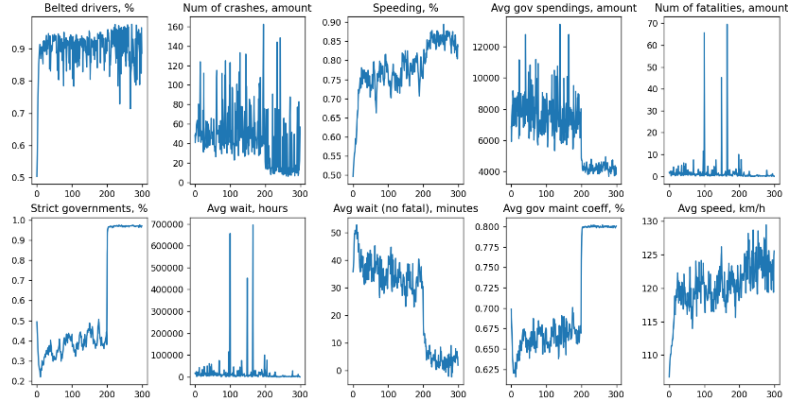


Fig. 6. Dynamics graphs running with *strict_g_chance* = 0.75 after the 200th epoch

4.3 Calculating effect maps

Now we will conduct a series of experiments to identify the impact of a combination of factors, with others fixed, on the average speed, the percentage of drivers wearing seat belts, the percentage of drivers speeding, and the average government spending. Within the framework of the study, we considered a number of combinations, but within the framework of the article we will consider the most notable ones, namely a pair of coefficients of distance reduction and accidents (see Figure 7) and a pair of coefficients of infrastructure degradation and duration of inspection (see Figure 8).

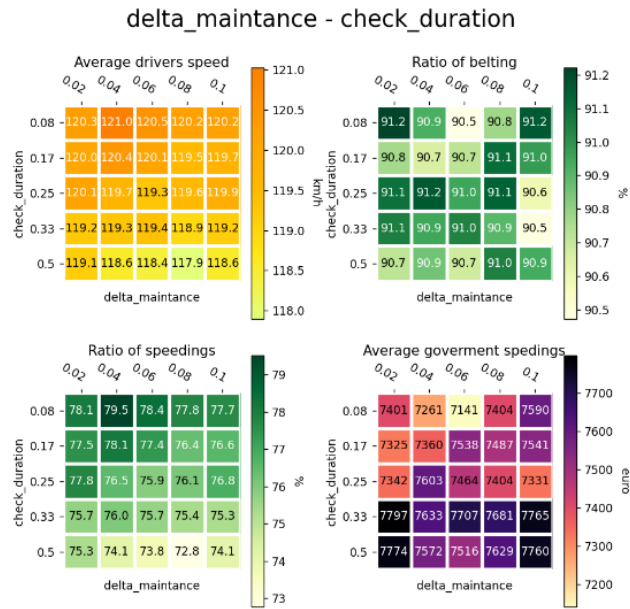


Fig. 7. Map of effects for *delta_maintenance* and *check_duration* coefficients

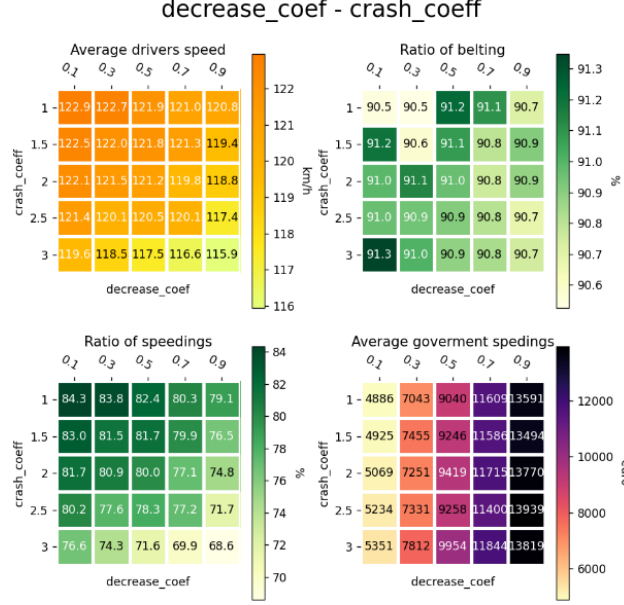


Fig. 8. Map of effects for crash and decrease coefficients

5 Conclusion and future work

As part of the experiments with the model, we obtained a number of interesting results. For example, taking into account the parameters of the national specifics of Germany, on average there is a slow trend towards stricter forms of regulation. While the average rate of driving with a belt is around 90%, speeding increased until the 24th epoch, after which it was fixed at an average of 75%.

Sustainability testing has shown that with the abrupt transition to a purely strict policy after the 200th epoch, there has been a sharp jump in the number of speeding drivers, but also an average reduction in travel time by 20 minutes. On the other hand, with a sharp transition to a mild form of regulation, the general trend towards an increase in the number of excesses also remains, but a smoother growth is observed. It is noteworthy that after a sharp transition to a purely strict model of regulation, government spends (on average about 4,154 euros per driver) less than in mild (on average 9,154 euros) and mixed (on average 7,485 euros) models.

A studying the effect of penalties amount on the frequency of offenses showed that in the framework of statistical testing, only the *fee_s_max_l* factor had an effect on the frequency of offenses. At the same time, its combination with the *fee_s_max_s* effect did not produce results, which partially confirms the early studies [6]. Also, using the example of effect maps, we see that increasing separately of *decrease_coef* and *check_duration* reduces the frequency of exceeding the speed limit, but increases the average government spending.

It is planned to test the approach in other countries in the future. Moreover, it is planned to improve the quality of the model by increasing the number of factors affecting the driver's driving speed, which in turn will lead to additional balancing indicators. Also, in the future, it would be interesting to consider the impact of the cost of an accident on the dynamics of key parameters.

Acknowledgements. The research was supported by The Russian Science Foundation, agreement №24-11-00272, <https://rscf.ru/project/24-11-00272/>.

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